

Job Application Tracker API

Role: Software Developer

Difficulty: junior

Project Type: fullstack

Duration: 14 days

Focus: api_development

Candidate: Rahul Sharma

Project Overview

Build a RESTful API to manage job applications with JWT auth and role-based access.

Technologies

- JavaScript
- TypeScript
- React
- Next.js
- Node.js
- Express.js
- Laravel (PHP)
- Tailwind CSS
- MongoDB
- JWT
- REST API design

Requirements

- Use existing skills, add JWT auth, CRUD, role-based access

Objectives

- Implement auth
- Create CRUD for applications
- Secure endpoints
- Document API
- Deploy locally

Functional Requirements

- User registration/login
- Create/read/update/delete applications
- Assign roles (applicant, recruiter)
- Filter applications by status
- Export application to PDF
- Rate limiting

Non-Functional Requirements

- JWT token expiry 1h
- Password hashing
- Input validation
- Error handling
- Unit tests

Modules

Auth

User auth and role management

- Register
- Login
- JWT issuance

Applications

CRUD for job applications

- Create
- Read
- Update
- Delete
- Filter

Users

User profile and role assignment

- Profile view
- Role update

Docs

API documentation

- Swagger UI
- Endpoint list

Implementation Plan

Setup

- Initialize repo
- Configure Node/Express
- Set up MongoDB

Auth

- Create user model
- Implement register/login
- Add JWT middleware

Applications

- Define schema
- Build CRUD routes
- Add role checks

Docs

- Add Swagger
- Document endpoints

Testing

- Write unit tests
- Run coverage

Deployment

- Configure .env
- Run locally

Evaluation Criteria

Auth Implementation (20%)

Correct JWT flow and role checks

CRUD Functionality (20%)

All CRUD operations work

Security (20%)

Password hashing, input validation

Documentation (20%)

Complete Swagger docs

Testing (20%)

Unit tests cover 80%

Deliverables

- API source code
- Swagger docs
- Postman collection
- Test report

Suggested Folder Structure

- src
- src/controllers
- src/models
- src/routes
- src/middleware
- src/docs
- tests

API Endpoints

POST /api/auth/register

Create user

POST /api/auth/login

Login user

GET /api/applications

List applications

POST /api/applications

Create application

GET /api/applications/:id

Get application

PUT /api/applications/:id

Update application

DELETE /api/applications/:id

Delete application

GET /api/docs

Swagger UI

Database Design

users

Store user data

- _id
- email
- passwordHash
- role

applications

Store job applications

- _id
- userId
- company
- position
- status
- createdAt

Testing Plan

- Unit tests for auth
- Unit tests for CRUD
- Integration test for JWT
- Test input validation
- Test error handling

Candidate Fit

Rahul's skills in Node, Express, JWT, and MongoDB align well with this project, offering a balanced challenge.